

COURSE DESCRIPTION

Course number	0418-141	Course Title	Computer Programming I
Credit hours	4 (3:3:4)	Course Coordinator	Dr. Hazem Raafat
Compulsory	Yes	Prerequisite	None

Textbook

Title:	<i>C++ How To Program (Late Objects Version)</i>
Author:	<i>Paul Deitel and Harvey Deitel</i>
Publisher:	<i>Pearson, Latest Edition</i>

Major Topics Covered in the Course

	Topic	Hours
1	Introduction to Programming and Programming Languages, Compilers, IDEs, Input/Output.	4
2	Programming Basics: Variables, Types, Expressions, Assignment, Casting, and Program Style.	5
3	Arithmetic Expressions and Boolean Expressions.	6
4	Conditionals, Loops and problem solving.	7
5	Functions.	4
6	Recursion.	3
7	Arrays (one-dimensional, multi-dimensional) and References.	4
8	Basic Searching and Sorting Algorithms.	2
9	Pointers.	4
10	Midterm I and Midterm II	3
	Total	42

Laboratory work:

	Topic	Week
1.	Stages of a computer program and steps on using the compiler	1
2.	Proper variable declarations. Stages in writing a simple program. Common syntax errors in simple programs	2
3.	Boolean Expressions and Conditional Statements (if, if-else if, nested if) I	3
4.	Boolean Expressions and Conditional Statements (if, if-else if, nested if) II	4
5.	Loops (while and for loop statements)	5
6.	Do-while, switch statements	6
7.	Nested loops	7
8.	Functions	8
9.	Recursion	9
10.	Parameter passing by reference, global variables, scope rules	10
11.	Defining and using one-dimensional arrays	11
12.	Multi-dimensional arrays	12
13.	Searching and sorting	13
14.	Pointers.	14

Grading

20%	Midterm 1
20%	Midterm 2
15-20%	Lab
0-5%	Hmk/ quiz
40%	Final

Course Outcomes

Students should be able to:

1. Analyze and explain the behavior of simple programs involving all constructs covered in this course.
2. Understand and implement basic concepts in programming including types, variables, scalar-vs-aggregated values, and parameter passing.
3. Design, implement, test, and debug programs that use the following constructs: arithmetic expressions, conditional statements and loops, one- and two-dimensional arrays, pointers, and recursion.
4. Decompose programs into well-defined functions.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	CS SOs					
	1	2	3	4	5	6
1.	H					
2.		H				
3.		H				
4.		H				

Last updated on Dec 8, 2019